

FPP Remote pHAT for Raspberry Pi Zero 2 W: 4–port RS–422 (Falcon differential) pixel driver, rev E (65 x 30 mm, same outline as the Pi Zero)

Pi mounts under this board on 11 mm standoffs (M2.5 holes H1–H4, 58 x 23 mm). Power: 12 V in on J4 -> F1 -> D1 -> U4 -> 5 V onto Pi pins 2/4.
NEVER power the Pi from its USB while 12 V is connected (a Pi Zero has no diode between its USB 5 V input and the header 5 V pins).
FPP: cape EEPROM U3 on i2c–1 @0x50 (fppcapedetect registers a 24c256). DPIPixels strings: port1=P1–29, port2=P1–31, port3=P1–26, port4=P1–7.
RJ45: port1 = pins 1(+),2(–) port2 = 3(+),6(–) port3 = 5(+),4(–) port4 = 7(+),8(–). Y = +, Z = –. Driver enables G and –G tied low.
JP1 open (default): EEPROM writable so FPP can program and sign it in place. Close JP1 afterwards to write–protect (WP high).

